



COURSE: ECO 206 STATISTICS II

NUMBER OF UNITS: 2 Units

INSTRUCTION: Attempt All Questions

TIME ALLOWED: 1 Hour

[20 Marks]

1. Considering that the GTI manufacturing industry is limited, on many occasions during the year, labour hours are lost because the machinery breaks down. The manager knowing from previous experience that the time-lost population is normally distributed, decides to keep track of the time lost during a randomly chosen week and calculates the mean time lost during 10 breakdowns as 820 hours, with a standard deviation of 510 hours. A 99% confidence interval is to be constructed for the mean time lost during the year's breakdowns. Also, determine the standard error and professionally interpret your estimates.

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} \quad 3 \text{ marks}$$

$$\begin{aligned} \bar{x} &= 820 \\ n &= 10 \\ s &= 510 \\ \alpha &= 0.01 \\ B &= 99\% \end{aligned}$$

2. Consider a Gold mining company in Zamfara State, North-West Nigeria. The firm is about to mine for gold deposits. The firm's geologists stated their subjective probabilities of finding gold as follows: P(gold) as 0.75 and P (no gold) as 0.25. The firm, after mining 200 feet, found no gold yet, but sampled the soil content. From experience, they knew the conditional probabilities of finding gold in this type of soil, given gold or no gold; P (this type of soil/gold) = 35%; P (this type of soil/no gold) = 65%.

- Determine the prior probabilities
- State the new information probabilities
- Estimate the Bayesian posterior probability of finding gold and suggest whether the firm should continue mining or not.

$$\frac{P(E) \times P(E|E)}{P(E) \times P(E|E) + P(E) \times P(E|E)} \quad 0.5 \text{ mark}$$

0.5 mark

2 marks

3. Table 1 below shows the probabilities associated with rolling two fair dice simultaneously. Use the information therein to provide answers to the questions that follow.

Table 1: Probability Density Function

x	0	1	2	3	4	5	6	7	8	9	10	11
P(x)	0.05	0.10	0.15	0.17	0.19	0.09	0.08	0.06	0.06	0.02	0.02	0.01

- Compute the expected value of the probability distribution 2 marks
- Estimate the variance of the probability distribution 2 marks
- Calculate the standard deviation and coefficient of variation of the distribution. 2 marks

4. With the aid of suitable diagrams, determine the following positions on the Gaussian Probability Distribution

- Find the area between $Z = -1.96$ and -1.56 . 2 marks
- Find the area between $Z = -2.34$ and 1.67 . 2 marks
- Find the value of Z , which is such that the area to its right is 0.2000. 2 marks
- Find the area between $Z = 0.60$ and 1.80 . 2 marks

0.0001

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NUMBER OF UNITS: 2 Units

INSTRUCTION: Attempt Questions One and Any Two Others

TIME ALLOWED: 2 Hours

Question One [25 Marks] Compulsory

One area of economic and business application of Statistics is investment decisions based on empirical data analysis. A Multinational Corporation (MNC) has reached out to PayStan, a consulting firm for investment advice, where you are currently on an internship. The MNC is interested in committing long-term funds to the Nigerian financial industry and wants to evaluate the earnings per share among four commercial banks: Zenith, Access, GT Bank, and First Bank of Nigeria (FBN), to inform its choice of investment destination. The CEO of the consulting firm, Dr Stanley Nwani, has provided the following live data, as presented in Table 1 below and requires the following output.

Table 1: Earnings per share (kobo)

	Zenith	Access	GTB	FBN
2020	630	225	0	94
2021	743	314	28	36
2022	747	469	301	54
2023	1,897	1,508	362	42
2024	2,979	1,241	1,223	73

Source: Annual Financial Reports, 2024

Requirements:

1a. Determine the sample statistic of the financial sector's grand average earnings per share for the period under consideration. [4 marks]

1b. Identify and justify the most suitable statistical technique to analyse the difference in the banks' average earnings per share. [2 marks]

1c. Conduct a test of hypothesis to validate either the null or alternative hypothesis:

$H_0: \mu_{Zenith} = \mu_{Access} = \mu_{GTB} = \mu_{FBN}$

$H_1: \text{The average earnings per share of the banks are significantly different.}$ [8 marks]

1d. Based on your findings above, what are your investment recommendations to the CEO of PayStan for the MNC's consideration? [4 marks]

1e. Waiting time has become a topical issue for the Ministry of Health, as patients continue to complain about high waiting times at Federal Medical Centres across Nigeria. To address patients' complaints on waiting time at FMC, the Medical Director of FMC, Asaba, wishes to know the average time patients must wait for healthcare services. The population of waiting times is known to be normally distributed, with a standard deviation of 16 minutes. A 99% confidence interval of the mean waiting time for 1,000 patients is to be constructed with the help of a simple random sample of 64 patients. The sample mean turns out to be 50 minutes. Determine and interpret the error bound of the mean and construct the confidence interval estimates. [7 marks]

Question Two [20 Marks]

2a. The chance of an IRS audit for a tax return with over ₦25,000 in income is about 2% per year. We are interested in the expected number of audits a person with that income experiences in 20 years. Assume each year is independent.

i. List the values that the random variable X could assume. [2 marks]

ii. How many audits are expected in 20 years? [2 marks]

iii. Find the probability that a person is not audited at all. [3 marks]

iv. Find the probability that a person is audited more than once. [4 marks]

v. Estimate the coefficient of variation of the probability distribution. [4 marks]

2b. A grinding machine is set such that its production of shafts has an average diameter of 10.10 cm and a standard deviation of 0.20 cm. The product specifications call for shaft diameters between 10.05 cm and 10.20 cm. What proportion of output meets the specifications, presuming a Gaussian distribution? [5 marks]

Question Three [20 Marks] ✓

3a. The building blocks of inferential statistics are sampling and probability distributions, which help statisticians select the best sample statistics (estimators) in inferring the unknown population parameters. From the foregoing statistical assertion, state the expected properties of a good estimator. [4 marks]

3b. Three Fintechs in Nigeria, Opay, PiggyVest, and Moniepoint, are competing for market share through strategic innovation and product development. A study of these Fintechs revealed that Opay plans to develop a new financial product to improve its market share from the current 20%. You have been hired by one of their competitors to study, analyse and recommend policy in reaction to Opay's plan. Given that Opay surveyed 600 customers, and their responses show that 180, 180 and 240 indicated preferences for Opay's new product, PiggyVest's existing product, and Moniepoint's existing products, respectively.

Required:

- At a 1% significance level, carry out a Chi-Square Analysis (following the chronological procedure) to show if the new product design would change the current market share among the Fintech firms, given that PiggyVest and Moniepoint have market shares of 35% and 45%, respectively. [8 marks]
- Estimate the mean, variance and standard deviation of the distribution. [3 marks]
- Based on your estimates in 3bi above, provide brief corporate strategic actions for Moniepoint, PiggyVest and Opay respectively. [5 marks]

Question Four [20 marks]

4a. Consider a Gold mining company in Zamfara State, North-West Nigeria. The firm is about to mine for gold deposits. The firm's geologists stated their subjective probabilities of finding gold as follows: $P(\text{gold})$ as 60%. The firm, after mining 3000 feet, found no gold yet, but sampled the soil content. From experience, they knew the conditional probabilities of finding gold in this type of soil, given gold or no gold; $P(\text{this type of soil}|\text{gold}) = 0.50$.

- Determine the prior probabilities [3 marks]
- State the new information probabilities [3 marks]
- Estimate the Bayesian posterior probability of finding gold and suggest whether the firm should continue mining or not. [6 marks]

4b. Write short notes with examples, formulae and area of applications where applicable on the following statistical concepts: i. Poisson Distribution ii. Joint Probability iii. Type I and Type II errors iv. Properties of a Normal Curve. [8 marks]

Question Five [20 Marks]

5a. If the income of 10,000 Nigerians is found to be normally distributed with a mean of ₦750 and a standard deviation of ₦50. What proportion of Nigerians have incomes exceeding ₦668 and ₦832, respectively? [6 marks]

5b. Consider GIG motors. Each month, thousands of buses arrive at the Ajah terminal. A 99% confidence interval for the proportion of all buses arriving on schedule will be estimated using a simple random sampling of 25 buses. The sample proportion of correct arrivals is 0.64. Also, determine the error bound of the proportion. Should the industry proportion of arrival time be 92%, are GIG motors time-efficient? [6 marks]

5c. Table 2 below shows the probabilities of rolling two fair dice. Estimate the expected values, standard deviation and coefficient of variation of the probability distribution. [7 marks]

x	0	1	2	3	4	5	6	7	8	9	10	11
P(x)	0.05	0.10	0.15	0.17	0.19	0.09	0.08	0.06	0.06	0.02	0.02	0.01